

Arnav Vasudev

(607)-375-7960 | av432@cornell.edu | US Citizen | [GitHub](#) | [LinkedIn](#)

EDUCATION

Cornell University

Bachelors of Science in Computer Science; **GPA: 3.9**

Aug 2022 – Dec 2025

Undergraduate coursework: Object Oriented Programming, Algorithms, Machine Learning, Deep learning, Linear Algebra, Numerical Analysis, Operating Systems

Graduate coursework: Distributed Systems, Reinforcement Learning, Formal Methods for Robotics

EXPERIENCE

Portal Lab | Research Intern

Jun 2024 – Present

Topic: Partially Observable Reinforcement Learning

- **Publication:** “Distilling Realizable Students from Unrealizable Teachers” IROS 2025.[\[Paper\]](#)
- Developed an algorithm for efficiently training end-to-end robot policies in partially observable settings, using a sim2real approach and teacher-student distillation.
- Trained an oracle reinforcement learning (PPO, SAC) policy with privileged simulation data in Mujoco MJX using PyTorch, then distilled it into a vision-based policy with point clouds.
- Transferred and deployed the policy to a real robot using ROS2, achieving reliable performance on manipulation tasks.

Tonbo Imaging | Software Intern (Computer Vision)

Jun 2025 – August 2025

Topic: Low-Light Image Enhancement

- Developed a low-light image enhancement model using PyTorch, implementing a U-Net architecture that transforms short-exposure images to match long-exposure quality for real-time deployment on memory-constrained devices.
- Deployed model at 30fps on a 4GB Jetson Nano using TensorRT with INT8 quantization and model distillation for memory optimization.
- Achieved PSNR of 41dB and SSIM 0.93 on 0.01 lux test images (moonlight conditions), demonstrating effective noise reduction and detail preservation in low-light scenarios.

TATA Capital | Data Science Intern

Jun 2023 – Aug 2023

Topic: Machine Learning for Financial Services

- Built targeted marketing models for loan product recommendation using ensemble methods on large-scale consumer financial datasets.
- Used Spark, SQL, Azure to engineer a data pipeline for processing millions of financial records across heterogeneous data sources.
- Deployed a classification model using PyTorch and XGBoost ensemble methods, achieving a 25% increase in customer segmentation.

PROJECTS

Autonomous RC car | [Website](#)

- Transformed an RC car into an autonomous vehicle by integrating an Artemis Nano, motor drivers, ToF sensors, and an IMU for real-time navigation. Tuned a PID controller for trajectory following and obstacle avoidance.

Tracking-by-detection | [GitHub](#)

- Finetuned a YOLO based neural network and implemented Kalman filtering for multi-object tracking to maintain player identities across frames, handling occlusions and predicting player trajectories for real-time tracking.

SKILLS

Languages: Python, C, C++, Java, OCaml, SQL, React JS, C#

Technologies: PyTorch, TensorRT, Isaac Gym, HuggingFace, CUDA, ROS, OpenCV, Open3D, Docker, Linux, GCP, SQL, Hadoop, Git, MongoDB, Spark, Azure, Kubernetes